

1. Natural Disasters May Have Played a Certain Role

The debate regarding how the climate of the Han Dynasty compares to contemporary times has no definitive answer to this day. Initially, Mr. Zhu Kezhen (竺可桢) believed that the Western Han belonged to a warm period, with the climate only beginning to turn cold during the Eastern Han. However, many scholars later questioned this, arguing that the Western Han actually existed in a relatively cold period, with temperatures likely not much different from today.

Regardless, the *Book of Han* (《汉书》) indeed records several natural disaster events caused by severe cold during the reign of Emperor Wu (occurring in the 4th year of Yuanguang, the 1st year of Yuanshuo, and the 2nd and 3rd years of Yuanding), proving that the climate back then was far from mild.

Yet, the even more severe disasters were floods. During the 54 years of Emperor Wu's rule, the Yellow River flooded and breached its dikes, causing disasters to the extent that it partially changed course twice—a situation that persisted for over a generation. In the 3rd year of Jianyuan, *"the river waters overflowed onto the plains, causing a great famine where people ate each other"* (河水溢于平原, 大饥, 人相食). In the 3rd year of Yuanguang, *"in the spring of the third year, the river waters shifted, flowing southeast from Dunqiu into the Bohai Sea"* (三年春, 河水徙, 从顿丘东南流入渤海), and *"the river waters breached at Puyang, inundating sixteen commanderies"* (河水决濮阳, 泛郡十六). In the 2nd year of Yuanfeng, the breach at Huzi was blocked, but *"after blocking the Xuanfang breach, the river again breached northward at Guantao, splitting into the Tunshi River"* (自塞宣房后, 河复北决于馆陶, 分为屯氏河).

Unlike the final millennium of the empire, when the economic center had already shifted to Southern China, the unprecedented flooding of the Yellow River—unseen since the time of Yu the Great—during the Western Han period directly struck the most populous and economically developed regions of the empire. It caused monumental destruction to social production. Therefore, the pressure brought to the Western Han court by frequent floods vastly exceeded the concerns of the monarchs in the Song, Yuan, Ming, and Qing dynasties, who worried more about refugees threatening political security or whether canal transport might be interrupted.

In fact, we can easily observe that to cope with the staggering destruction of the floods, the empire adopted a series of unprecedented measures, which demanded a historically unprecedented strengthening of the state's administrative power. First, controlling the floods inherently required conscripting massive numbers of soldiers to fill the breaches. To deal with the Yellow River breach in the 3rd year of Yuanguang, 100,000 soldiers were mobilized. When Emperor Wu personally visited the Yellow River breach in the 2nd year of Yuanfeng to ultimately block the breach at Huzi, tens of thousands of soldiers alone were deployed.

Secondly, the floods caused massive numbers of people in the Guandong region to become destitute, homeless, and devoid of food and clothing. The number of refugees was colossally massive relative to the Han Empire's total population of over 30 million. The highest recorded figure was in the 4th year of Yuanfeng, noting *"2 million refugees in Guandong, and 400,000 without registered names/numbers"* (关东流民 200 万口, 无名数者 40 万) (*Biographies of the Wan Shi and Zhang Shu in the Book of Han* / 《万石张叔列传》). In the 4th year of Yuanshou, there were also over 700,000 refugees in Guandong (*Treatise on the Balanced Standard* / 《平准书》).

To cope with disasters of such intensity, the empire, on one hand, needed to requisition grain from relatively abundant regions to provide relief to the disaster zones. For example, in the autumn of the 3rd year of Yuanding (September), because the Jiangnan region suffered from floods, Emperor Wu worried that the people would be unable to survive the winter hunger and cold, and thus dispatched grain from the Bashu region for disaster relief (*Basic Annals of Emperor Wu* / 《武帝本纪》).

On the other hand, it was necessary to arrange for these refugees to migrate to places with arable land. The destinations were mostly frontier commanderies in the empire's northwest, such as Longxi, Beidi, and Shangjun, as well as newly conquered territories, thereby simultaneously achieving the goal of consolidating border defenses (实边). The sheer scale of migration during Emperor Wu's reign was staggering. In the winter of the 4th year of Yuanshou, impoverished people from Guandong relocated to Longxi, Beidi, Xihe, Shangjun, and Kuaiji totaled 725,000 individuals (关东贫民迁徙到陇西、北地、西河、上郡、会稽总共 725000 口) (*Basic Annals of Emperor Wu* / 《武帝本纪》).

During the Yuanfeng era, when the expansionary wars were at their most intense, the empire further "newly established the Zhangye and Jiuquan commanderies, while agricultural officials were set up in Shangjun, Shuofang, Xihe, and Hexi, deploying 600,000 border-defense soldiers to garrison and farm there" (初置张掖、酒泉郡，而上郡、朔方、西河、河西开田官，斥塞卒 60 万戍田之) (*Treatise on the Balanced Standard* / 《平准书》). Earlier, in the summer of the 2nd year of Yuanshuo, they also "recruited 100,000 people to relocate to Shuofang" (募民徙朔方十万口) (*Basic Annals* / 《本纪》). Migrations involving fewer than 100,000 people are not counted here due to scattered records (such as those in the *Biographies of the Xiongnu* / 《匈奴列传》).

Because the unified Han Empire vastly exceeded the vassal states of the Warring States period in both population and territory, the difficulty and cost of dispensing grain and relocating people were absolutely incomparable to what King Hui of Liang did back then when he "moved his people [or grain] to Hedong, and moved his grain [or people] to Henei" (移其民（粟）于河东，移其粟（民）于河内).

Even more notably, relocating the refugees from Guandong into the newly conquered commanderies and counties in the northwest meant their route would pass right through the vital capital region of Guanzhong (关中京畿重地). Consequently, the court had to invest considerable effort into political security. The *Treatise on the Balanced Standard* (《平准书》) records: "Emissaries divided them into sections to escort them, with official carriages in continuous sight" (使者分部护之，冠盖相望), and ultimately, "the expenses were counted in the hundreds of millions and were beyond calculation. Consequently, the state treasury became utterly depleted" (其费以亿记，不可胜数。于是县官大空).

At other times, the migration of people was combined with the transportation of grain. The *Treatise on the Balanced Standard* also records that when the region east of the mountains (Shandong/Guandong) suffered severe disasters—where people cannibalized each other across an area of one to two thousand *li*—Emperor Wu issued an edict allowing the starving masses to travel south toward the Jianghuai region to seek sustenance. If they wished to remain locally, they could do so, while the court simultaneously dispatched grain from the Bashu region in the upper reaches of the Yangtze River for disaster relief.

Thirdly, to cope with the colossal destruction inflicted by natural disasters on social production, the empire's emphasis on agricultural production similarly reached an unprecedented level. For example, in the 3rd year of Yuanshou, the court "*dispatched emissaries to instruct the commanderies that had suffered floods to plant winter wheat*" (遣谒者却有水灾郡种宿麦) (*Basic Annals* / 《本纪》).

During Emperor Wu's reign, the majority of the water conservancy projects of both Han dynasties were completed. The court explored the latest technologies such as the *Jingqu* method (井渠法 / karez well-canal system), dispatched court officials to improve farming methods, popularized the production of crops beneficial for famine preparation (winter wheat and soybeans), and vastly improved agricultural tools. Even when compiling the new calendar during the Taichu era, the solar terms (节气) were concurrently established to guide agricultural production.

Given the incredibly high expectations people at the time placed on improving agricultural efficiency, it is hard not to view this as a reaction to the mercilessness of natural disasters. And these government-led missions demanded the expansion of state power.

Finally, precisely because of the frequency and uncertainty of natural disasters, the court became increasingly inclined to strengthen macroeconomic control to hedge against the severe impacts of famine years (especially the loss of independent peasant-proprietors caused by land annexation). Once this combined with the ballooning fiscal expenditures brought about by wars and disaster relief, the *Pingzhun Junshu* (平准均输 / Equitable Delivery and Balanced Standard system) was born; it even became the very symbol of the expansion of state power in the Han Dynasty. Aside from floods, the records concerning droughts during Emperor Wu's reign are even more numerous than those of floods (though this does not mean droughts were more severe than floods, as the floods were practically in a continuous state because the Yellow River breaches remained unblocked), and locust plagues were also highly common.

Today, we can still discern from Han dynasty historical records the intense psychological reactions of the people back then to these frequent and severe natural disasters. During the Western Han period, there existed an Ancestral Temple System (宗庙制度). Beginning with Emperor Wen, the temples of successive emperors each had a specific title (a sort of name). For instance, Emperor Wen's temple was named *Gucheng* (顾成 - Fulfilling Success), Emperor Jing's was *Deyang* (德阳 - Radiant Virtue), Emperor Xuan's was *Leyou* (乐游 - Joyous Excursion), and Emperor Yuan's was *Changshou* (长寿 - Longevity). All of these were names signifying auspiciousness, peace, and joy.

However, Emperor Wu's temple was named "**Longyuan**" (龙渊 - Dragon Abyss). The *Basic Annals of Emperor Wu* (《武帝纪》) links the construction of the Longyuan Palace with the Yellow River breaching at Puyang in the 4th year of Yuanguang, for which 100,000 soldiers were conscripted to block the breach.

Although today people still cannot know for certain exactly how many "Longyuan Palaces" existed back then—whether there was only the temple of Emperor Wu in the Guanzhong region as recorded in the *Sanfu Huangtu* (《三辅黄图》 / *Illustrated Guide to the Three Capital Regions*), or whether there was another Longyuan Palace near the Yellow River breach intended to pray for the subsidence of the floods, much like the later Xuanfang Palace (宣房宫).

Nevertheless, my judgment remains that, whether or not there was a "Longyuan Palace" beside the Yellow River in Henan, the naming and construction of "Longyuan Palace" are highly unlikely to be unrelated to the flooding and breaching of the Yellow River.

The first reason is that when Ban Gu (班固) compiled the *Book of Han*, it had only been two centuries since Emperor Wu's time. Massive remnants of Western Han architecture still existed then, making it not so easy to mistakenly link two entirely unrelated events. The second reason is that the recorded temple names of other Western Han emperors mostly carried images of auspiciousness and blessings. It is highly unlikely that Emperor Wu's temple alone was named after some mere decorative ornament (i.e., a "bronze flying dragon").

Therefore, I believe that the name "Longyuan" should have served the exact same function as the temple names of other Western Han emperors: praying for blessings. It is simply that what Emperor Wu considered here was not his personal fortune, but the hope that the Yellow River floods would finally subside.

The *Treatise on the Feng and Shan Sacrifices* (《封禅书》) records that when Emperor Wu heavily favored the alchemist (方士) Luan Da (栾大), Luan Da fraudulently claimed that his master had once said: "Gold can be transmuted, the Yellow River breaches can be blocked, the elixir of immortality can be obtained, and one can attain the Dao and ascend as an immortal" (黄金可以变, 黄河决口可以堵塞, 不死之药可以获得, 也可以得道升仙).

However, Emperor Wu—a monarch who has historically been viewed as deeply superstitious regarding alchemy and occult arts—made no mention of alchemy, elixirs of immortality, or methods for ascending to immortality in his edict enfeoffing Luan Da as the Marquis of Letong (乐通侯). He only left behind a record expressing immense anxiety over the fact that the Yellow River was experiencing such ceaseless and severe flooding for the first time since the flood-taming of Yu the Great, harboring the hope that an extraordinary figure like Luan Da could genuinely resolve this crisis.

Of course, this is not to say that Emperor Wu merely pinned his hopes on alchemists when facing natural disasters. We already know that he indeed adopted measures unprecedented by his predecessors, and the results were obvious: despite such severe internal woes, the empire achieved strategic dominance in the war against the Xiongnu through massive and protracted campaigns, doubling the size of its territory. But Emperor Wu was, after all, a man from over two millennia ago. Even people today turn to religion for psychological comfort when facing uncontrollable problems; it was perfectly normal for someone back then—even a hero—to seek psychological solace from the unknowable.

The intellectuals of the empire were equally profoundly concerned about the disasters. Sima Qian (司马迁), the author of the *Records of the Grand Historian* (《史记》), is a quintessential representative. Although he personally disagreed with many of the empire's policies, his sincere yearning for a peaceful realm still radiates between the lines.

The *Treatise on Rivers and Canals* (《河渠书》), as one of the Eight Treatises of the *Records of the Grand Historian*, concentratively chronicles the immense efforts people back then invested into combating floods. Even after the tragic event that occurred between him and the emperor in his later years, when Sima Qian recorded the felling of the bamboo in the Qi Garden (伐淇园竹) to block the Xuanfang breach—an event he personally experienced while following the Feng and Shan sacrifice

procession—he adopted a highly praiseful and nostalgic tone. He believed that although Emperor Wu's reign saw countless investments and numerous accomplishments in water conservancy, "*yet the most prominent among them was at Xuanfang*" (然其著者在宣房), and he completely transcribed the *Song of Huzi* (《瓠子歌》) composed by Emperor Wu.

He stated bluntly: "*I followed to carry firewood on my back to block the Xuanfang breach, and mourning the Song of Huzi, I composed the Treatise on Rivers and Canals*" (余从负薪塞宣房，悲瓠子之诗而作河渠书). The *Treatise on Rivers and Canals* thus became the only chapter in the *Records of the Grand Historian*—other than the Grand Historian's own preface—where he explicitly explained why he wrote it. One can well imagine the status that flood control held in Sima Qian's heart.

"*Those in power vied to speak of water conservancy*" (用事者争言水利). This did not only refer to intellectuals serving as officials in the court or localities; it equally included intellectuals out of office. The *Treatise on Ditches and Moats* in the *Book of Han* (《汉书 沟洫志》) records exactly such a "*person offering convenient advice*" (言便宜者)—a man named Yan Nian (延年) from the Qi region. He submitted a memorial to Emperor Wu suggesting that, according to maps and topographic inspections, the Yellow River should be diverted to flow toward Mongolia, entering the sea eastward from a much more northern position. This way, Guandong would never suffer from floods again, and the North would not have to worry about the Xiongnu because the Yellow River waters would serve as a natural barrier. Consequently, countless expenditures of money and manpower could be saved.

Facing a suggestion that bordered on the fantastical even by the standards of that era, the emperor himself probably felt caught between tears and laughter. Yet, not wishing to crush such unadorned, patriotic sentiment concerning state affairs, he could only reject it by deflecting, stating that the Yellow River's channel was opened by Yu the Great, and he possessed no qualifications to alter it.

1, 自然灾害可能起到了一定作用

关于汉代气候与当代相比如何的争论，直到今天也没有确切答案。最初竺可桢先生认为西汉属于一个温暖期，从东汉才开始转冷。然而后来有很多学者质疑这一点，认为西汉实际上处于一种相对寒冷的时期，气温当与今相差不大。无论如何，《汉书》中确实记载了汉武帝统治时期数次由严寒引起的自然灾害事件（发生在元光四年，元朔元年，元鼎二年和元鼎三年），足见当年气候远非平和。而更为严重的灾害则是水灾，武帝统治的 54 年间黄河泛滥决口成灾，以至于发生两次部分改道，持续超过了一代之久。建元三年“河水溢于平原，大饥，人相食”，元光三年“三年春，河水徙，从顿丘东南流入渤海”“河水决濮阳，泛郡十六”，元封二年堵塞瓠子决口，然而“自塞宣房后，河复北决于馆陶，分为屯氏河”。与帝国最后一千年经济重心已经转移到中国南方不同，西汉时期黄河自大禹之后的空前泛滥直接打击到了帝国人口和经济最发达的地区，对社会生产造成了极大破坏，因此水患频仍给西汉朝廷带来的压力远远大于更多担心泛区流民威胁政治安全，运河漕运是否可能因此中断的宋元明清的君主。事实上，我们可以很容易看到为了应对水灾如此惊人的破坏，帝国采取了一系列前所未有的措施，而这些措施则要求史无前例地加强国家的行政力量。首先，要想控制水患，本身就需要征发大量士卒填补决口，为应对元光三年的黄河决口征发了士卒十万，元封二年汉武帝亲临黄河决口处最终堵塞了瓠子一地的决口，仅士兵就投入了数万人。

其次，水患造成关东地区大量人口流离失所，缺衣少食，而流民的数量相对于汉帝国总共 3000 万余的人口极为庞大，其中记载最多的一次是元封四年，“关东流民 200 万口，无名数者 40 万”（《万石张

叔列传》），元狩四年关东流民也有 70 余万口（《平准书》）。为了应对如此烈度的灾害，帝国一方面需要从粮食相对富裕的地区征调粮食到灾区赈灾，例如元鼎三年秋九月，汉武帝因为江南地区遭受水灾而担忧一旦入冬人民饥寒无法生存，于是从巴蜀地区调粮赈灾（《武帝本纪》）。另一方面也需要安排这些流民迁徙到有田可耕的地方生活，而目的地大多为陇西、北地、上郡这样帝国西北方的边郡以及新征服的领土，进而同时达到实边的目的。汉武帝年间移民迁徙的数量极为庞大，元狩四年冬，关东贫民迁徙到陇西、北地、西河、上郡、会稽总共 725000 口（《武帝本纪》），在元封年间扩张战争最为剧烈的时候，又“初置张掖、酒泉郡，而上郡、朔方、西河、河西开田官，斥塞卒 60 万戍田之。”（《平准书》）更早的元朔二年夏也有“募民徙朔方十万口”（本纪），十万以下人数的移民因记载分散（例如在《匈奴列传》中）这里不统计。由于统一的汉帝国在人口和疆域上远远超过了战国时期的诸侯国，发粟徙民的难度以及成本绝非当年梁惠王“移其民（粟）于河东，移其粟（民）于河内”能够比拟的。

更值得注意的是，将关东的流民迁入西北新征服的郡县，从路线上说会经过关中京畿重地，因而，朝廷还不得不在政治安全上下相当功夫，《平准书》就记载“使者分部护之，冠盖相望”，最终“其费以亿记，不可胜数。于是县官大空”。还有的时候将移民与移粟相结合，《平准书》里还有山东地区受灾严重，方圆一二千里人或相食，汉武帝下诏让饥民南下江淮一带就食，如果想留在当地，就留在当地，同时从长江上游再次调巴蜀地区的粮食赈济的记录。

其三，为了应对自然灾害对社会生产造成的巨大破坏，帝国对农业生产的重视同样到了空前的地步，例如元狩三年“遣谒者却有水灾郡种宿麦”（本纪）。武帝一朝完成了两汉大部分的水利建设，探索了向井渠法这样的最新技术，派朝廷官员改良耕作方法，普及对于备荒有利的作物生产（冬小麦和大豆），生产工具也得到了极大改良。甚至在太初年间编订新历法的时候，也顺带确定了节气用来指导农业生产。当时的人们对提升农业生产效率给予了如此高的期待，很难不认为是面对天灾无情时的反应，而这些官府主导的任务则要求国家力量的扩大。最后，正是由于自然灾害的频繁和不确定性，朝廷为了对冲荒年产生的恶劣影响（特别是兼并带来的自耕农的流失），就更加倾向于加强宏观的调控。这一旦与战争和救灾带来的财政支出的膨胀相结合，平准均输就诞生了，它甚至成为了汉代国家权力扩大的象征。除了水灾以外，武帝年间有关旱灾的记录甚至比水灾更多（但这并不意味着旱灾比水灾严重，因为水灾实际上因为黄河决口没有堵住处于一种持续不断的状态），蝗灾也十分常见。

今天我们仍能从汉代的历史记载中看到当年人们对于频发和严重的自然灾害心理上的强烈反应。西汉时期存在一套宗庙制度，自文帝开始历代皇帝之庙各有号（一种名称），例如文帝庙名顾成，景帝庙名德阳，宣帝庙名乐游，元帝庙名长寿，这些均是祥瑞安乐的名称。汉武帝的庙名字叫“龙渊”，《武帝纪》中将龙渊宫的修建与元光四年黄河在濮阳决口，征发士卒十万以堵决口相联系。虽然时至今日人们仍然无法确切地知道当年究竟有几个龙渊宫，是只有《三辅黄图》里面记载的关中地区那个汉武帝之庙，还是在黄河决口附近如同后来的宣房宫一样另有一个意在祈祷水患平息的龙渊宫。不过我的判断仍然是，无论是在河南黄河边上有没有一个“龙渊宫”，“龙渊宫”的命名和修建都不太可能与黄河决口泛滥之事无关。第一个原因是班固修《汉书》时离武帝时期不过两个世纪，西汉时期的建筑在当时有大量遗存，没那么容易把两个毫不相关的事情错误地联系在一起。第二个原因是有记载的其他西汉皇帝的庙名多是祥瑞之象和祝福之意，也不太可能独独武帝的庙是以某个装饰品（即“铜飞龙”）来命名。因此我认为“龙渊”之名应当与西汉诸帝之庙名称具有相同的功用，那就是祈福，只不过汉武帝这里考虑的不是他个人，而是希望黄河水患能够平息。

《封禅书》中记载当年汉武帝重用方士栾大之时，栾大诈称自己的师傅曾经说过“黄金可以变，黄河决口可以堵塞，不死之药可以获得，也可以得道升仙”。然而，就是汉武帝这样一个自古都被视为迷信方术的君主，在封栾大为乐通侯的诏书中，没有谈到炼金术、不死药和修仙法，只留下了对于大禹治河以来黄河第一次出现这么严重的泛滥不止的情况表达了极大的忧虑，并希望像栾大这样的奇人异士真的能解决此事。当然，这里不是说汉武帝面对自然灾害仅仅寄希望于方士，我们已经知道他的确采取了前人所未做过的措施来应对，而且成果显而易见，帝国在如此内忧的情况下通过大规模的持久战争取得了对匈奴战争的战略优势，国土翻了一倍。但汉武帝毕竟是两千多年前的人，即使今天的人们在面临

难以控制的问题时也会求助于宗教，那个时候的人，哪怕是一个英雄，去对不可知的事物寻求心理慰藉也是非常正常的。

帝国的知识分子同样对于灾害极为关心，《史记》的作者司马迁就是其中的典型代表，尽管他本人并不赞同帝国的许多政策，但其希望天下无事的赤诚之心却依然流露在字里行间。《河渠书》作为《史记》八书之一集中记述了当年人们应对水患所付出的心血，即使在晚年与皇帝之间发生了那样悲剧性的事件，但司马迁在记述伐淇园竹，塞宣房这一他跟随封禅队伍亲身经历之事时，他采取了一种相当赞美和怀念的口吻，认为虽然武帝一朝在水利上投入无数，完成的事业众多，“然其著者在宣房”，并且完整地记录下了武帝所作《瓠子歌》。他直言道“余从负薪塞宣房，悲瓠子之诗而作河渠书。”，《河渠书》也成为史记中除了太史公的自序以外唯一交代为什么要写的篇目，水患治理在司马迁心中的地位可想而知。“用事者争言水利”不仅是说那些在朝廷或地方为官的知识分子们，同样也包括了在野的知识分子，《汉书 沟洫志》中记载了这样一位“言便宜者”，齐地一个叫延年的人给武帝上书说，应该按照地图并考察地形，让黄河改道流向蒙古，从更北的位置向东入海，这样关东就再不会有水灾，而在北方也不用担心匈奴，因为黄河水成为了自然的屏障，这样一来就可以省掉无数钱财和人力的开支。面对这样一个即使在当年看来都近乎异想天开的建议，皇帝本人大概也感到哭笑不得，但是又不想打击这种忧心国事的朴素情感，只能推脱于黄河的河道乃是大禹所开，自己没有资格去进行更改来驳回。